Supplemental Materials

Supply Dominance, Leak Identifiability and Service-Failure Prediction in Water Distribution Networks

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Propositions 1–4 and the experiment labels (X1, X2, R1–R3) refer to the main paper. Tables S1–S7 and Fig. S1 support Sections 3, 5 and 6.

Table S1: Novelty boundary of the structural results.

Source	Content used or derived
Graph theory (existing)	Dominators and their near-linear computation (Lengauer and Tarjan 1979); articulation points and biconnected components (Hopcroft and Tarjan 1973). Used to compute shadows.
Hydraulics (existing)	Unique steady state from the convex content formulation (Collins et al. 1978); leak sensitivity matrix and angle method (Pérez et al. 2011; Casillas et al. 2014); isolability-based sensor placement (Casillas et al. 2013). Used as tools and baselines.
Electrical-network theory (existing)	Harmonic functions and hitting probabilities of random walks on weighted graphs (Doyle and Snell 1984). Used in the proof of Proposition 2.
This paper (derived)	Exact head invariance in any shadow for leaks of any size outside it (Prop. 1); hitting-probability form of ρ with connected decay, and the interpolation corollary (Prop. 2, Cor. 1); reduction to a scalar response and observational equivalence for sensors in a shadow (Prop. 3); location of the concurrent-leak cross term and its propagation to a sign-indefinite network interaction (Prop. 4).

Table S2: Experiments, data and decision rules. All seeds and settings are fixed in the published code.

Experiment	Data (training / test)	Metric, uncertainty	Decision rule
Prototype	Synthetic and KY4, 150 leak sites, 1–10 cm^2, 30 random layouts (10%)	Spearman, R^2, superposition error (medians)	None
Placement	Selection library (1 cm^2, nominal demand); R2: independent library (5 cm^2, 1.2× demand), noise 0.01 and 0.05 m	Mean hop error; random layouts: mean and s.d. over 10	R2: even spreading beats shadow rule and PageRank at every budget
X1 localiser	Per network 3,000 training (seed 0) / 800 test (seed 1) scenarios; 30% two leaks, 1–10 cm^2, demand 0.6–1.4×, noise 0.01 m	Share within 300 m; t-interval over 5 seeds	Hybrid improves on prior and flags help if intervals above 0 on both networks
R3 localiser	As X1, hybrids retrained, per-scenario results kept	As X1 plus paired scenario bootstrap, McNemar, hierarchical bootstrap (seeds and scenarios, 2,000)	As X1, on the hierarchical interval
Service failure	300 scenarios per network, $A_0 = 2\,\mathrm{cm}^2$	Partial Spearman; median relative error; top-1	Disequilibrium: $p < 0.01$ on both; size: beats both baselines; node: beats all baselines
X2 failure size	Calibration 300 / test 300 per network (new seeds), $A_0 = 2$, $A_1 = 4\,\mathrm{cm}^2$	Median relative error, Spearman, share within 50%	Error < 0.5 and below LIN on both networks
R1 robustness	100 leak junctions per network × 7 conditions (paired)	As X2, plus noise and wrong-junction tests	Model route robust if error < 0.5 on both networks
R4, R5	X2 data; libraries; BattLeDIM results	Coefficients, learning curves, checks	None

Table S3: Worked example of Proposition 2 on the synthetic network: linearised ratio $\mathbf{G}_{ij}/\mathbf{G}_{jj}$, Monte Carlo hitting probability (MC) and simulated drop ratios at 1 and 10 cm^2, for selected junctions i and leak junctions j.

Leak j	Junction i	$\mathbf{G}_{ij}/\mathbf{G}_{jj}$	MC	Simulated, 1 cm^2	Simulated, 10 cm^2
G3_0 (entry)	G2_3	0.727	0.726	0.742	0.609
	G0_2	0.483	0.485	0.479	0.401
	D7 (in shadow)	1.000	1.000	1.000	1.000
G2_3 (grid)	G3_0	0.788	0.791	0.786	0.732
	D0, D7	0.788	0.786, 0.788	0.786	0.732
	G5_5	0.185	0.183	0.184	0.169
D3 (in branch)	D0	0.368	0.366	0.352	0.306
	G3_0	0.094	0.098	0.080	0.043
All 36 pairs: maximum absolute difference		–	0.007	0.025	0.118

Table S4: Signed interaction of concurrent 10 cm^2 leaks (30 pairs per type): median relative signed interaction $\sum_i \delta_i / \sum_i (\Delta h_i(a) + \Delta h_i(b))$, share of pairs with positive interaction, and share of the cross-term magnitude on supply, return and opposed links.

Network	Pairs	Leak model	Signed	Positive	Supply	Return	Opposed
Synthetic	Common split	Fixed outflow	−0.128	0%	0.11	0.80	0.09
		Orifice, PDD	−0.144	0%	0.11	0.80	0.09
	Random	Fixed outflow	−0.086	20%	0.16	0.71	0.13
		Orifice, PDD	−0.102	10%	0.16	0.71	0.13
KY4	Common split	Fixed outflow	+0.027	83%	0.56	0.36	0.08
		Orifice, PDD	+0.008	60%	0.53	0.38	0.08
	Random	Fixed outflow	+0.000	63%	0.41	0.23	0.37
		Orifice, PDD	+0.000	60%	0.38	0.24	0.38

Table S5: Per-seed share of estimates within 300 m in R3 (800 test scenarios) and McNemar tests of the hybrid against the hydraulic model.

	L-Town (model 0.750)		KY4 (model 0.464)	
Seed	Hybrid / without flags	McNemar p	Hybrid / without flags	McNemar p
0	0.786 / 0.812	0.034	0.445 / 0.426	0.319
1	0.766 / 0.810	0.393	0.415 / 0.411	0.0101
2	0.792 / 0.792	0.0176	0.405 / 0.410	0.00024
3	0.801 / 0.790	0.00269	0.379 / 0.440	1.0×10^{-6}
4	0.738 / 0.775	0.544	0.396 / 0.407	1.6×10^{-5}

Table S6: First service-failure experiment ($A_0 = 2\,\mathrm{cm}^2$, 300 scenarios per network).

Prediction	L-Town	KY4	Verdict
Partial Spearman of A^* with Ψ_N, given total	−0.734	−0.828	Holds on both
Spearman of A^* with total depletion	−0.100	−0.406	
A^* median relative error: LIN	**1.67**	**1.06**	
failure at the leak junction (AT-LEAK)	3.69	1.74	Holds on both
lowest margin (MARGIN-MIN)	∞	∞	
First failing junction, top-1: $\arg\max D_i$	0.520	**0.707**	
largest drop	**0.685**	0.364	Fails on L-Town
nearest junction	0.664	0.378	
lowest margin	0.034	0.058	

Table S7: Generalisation of REG: median relative error on the X2 test sets by calibration-set size (median over 200 random subsets), across networks and with a pooled fit.

	Calibration scenarios (same network)							
Network	10	20	40	80	160	All	Other network	Pooled
L-Town (2PT 0.107)	0.094	0.082	0.079	0.077	0.078	0.080	0.194	0.100
KY4 (2PT 0.223)	0.211	0.172	0.155	0.143	0.144	0.138	0.158	0.120

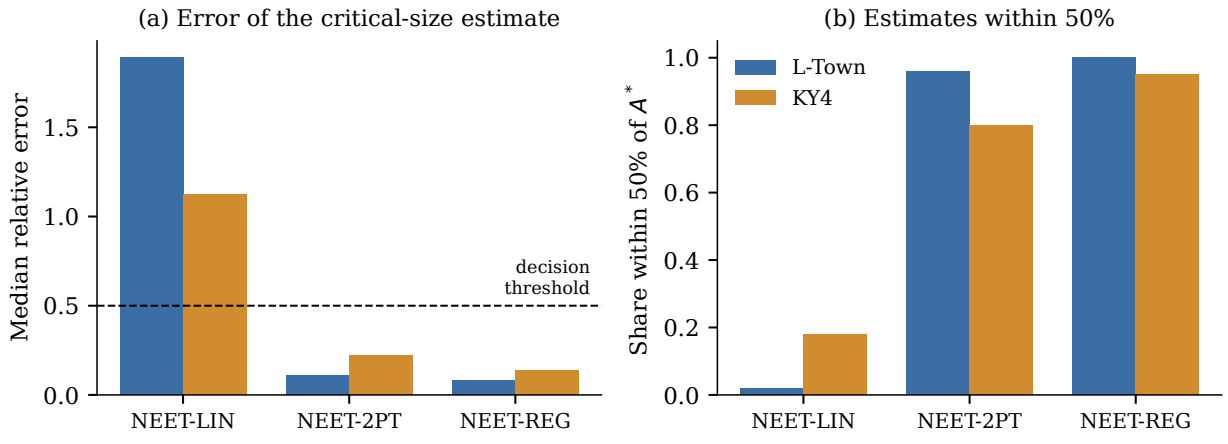


Figure S1: Failure-size experiment X2: (a) median relative error of the three estimators on held-out scenarios, with the pre-specified threshold of 0.5; and (b) share of estimates within 50% of the true critical area.